RACE
Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-01

Single Correct Question

1. A man has to travel from Bombay to Goa via Pune. He has 3 ways from Bombay to Pune & 4 ways from Pune to Goa. In how many ways he can plan a return journey from Bombay to Goa back to Bombay?
(A) 121 (B) 144 (C) 169 (D) 289
MPC177

2. How many squares are there on chess Board?
(A) 204 (B) 206 (C) 208 (D) 210
MPC178

3. There are three stations, A, B and C. Four routes for going from station A to station B and three routes for going from station B to station C. Find the number of different ways through which a person can go from A to C via B.
(A) 10 (B) 12 (C) 14 (D) 16
MPC179

4. A hall has 10 gates. In how many ways, can a man enter the hall through one gate and come out through a different gate.
(A) 70 (B) 80 (C) 90 (D) 100
MPC180

5. 10 students compete in a swimming race. In how many ways can they occupy the first 3 positions.
(A) 1024 (B) 528 (C) 720 (D) 529
MPC181

6. The number of ways in which first, second and third prizes can be given to 5 competitors is
(A) 10 (B) 60 (C) 15 (D) 125
MPC182

7. In how many ways can 12 true-false questions be replied
(A) 20 (B) 4096 (C) 2048 (D) 1024
MPC183

8. If a man and his wife enter in a bus, in which five seats are vacant, then the number of different ways in which they can be seated is
(A) 2 (B) 5 (C) 20 (D) 40
MPC184

9. Given 4 flags of different colours, how many different signals can be generated, if a signal requires the use of 2 flags one below the other?
(A) 10 (B) 11 (C) 12 (D) 13
MPC185

10. A coin is tossed 3 times and the outcomes are recorded. How many possible outcomes are there?
(A) 8 (B) 9 (C) 10 (D) 11
MPC186

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	A	B	C	C	B	B	C	C	A

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-02

Single Correct Question

1. Determine n if ${}^{2n}C_3 : {}^{n}C_3 = 12 : 1$
(A) 5 (B) 6 (C) 7 (D) 8
MPC187

2. The number of 3 digits odd numbers, that can be formed by using the digits 1, 2, 3, 4, 5, 6 when the repetition is allowed, is
(A) 60 (B) 108 (C) 36 (D) 30
MPC188

3. How many five digits numbers can be made by using a digit 0, 1, 2, 4, 5, 3 which are divisible by 3. (If repetition not allowed)
(A) 186 (B) 196 (C) 206 (D) 216
MPC189

4. How many Numbers can be made by using the digits 3, 4, 5, 6, 7 which are greater than 56000.
(A) 40 (B) 50 (C) 60 (D) 70
MPC190

5. How many four digits Even numbers can be made by using 0, 1, 3, 4, 7, 9
(Repetition not allowed)
(A) 98 (B) 108 (C) 118 (D) 128
MPC191

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6. How many four digits Odd numbers can be made by using 0, 1, 3, 4, 7, 9
(Repetition not allowed)
(A) 72 (B) 102 (C) 152 (D) 192
MPC192

7. How many of the 900 three digit numbers have at least one even digit?
(A) 775 (B) 875 (C) 450 (D) 750
MPC193

8. Number of natural numbers between 100 and 1000 such that at least one of their digits is 7, is
(A) 225 (B) 243 (C) 252 (D) None
MPC194

9. Number of 4 digit numbers of the form $N = abcd$ which satisfy following three conditions:
(i) $4000 \le N < 6000$ (ii) N is multiple of 5
(iii) $3 \le b < c \le 6$ is equal to
(A) 12 (B) 18 (C) 24 (D) 48
MPC195

10. A committee of 3 persons is to be constituted from a group of 2 men and 3 women. How many of these committees would consist of 1 man and 2 women?
(A) 3 (B) 4 (C) 5 (D) 6
MPC196

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	B	D	C	B	D	A	C	C	D

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Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-03

Single Correct Question

1. If the letters of the word $SACHIN$ arranged in all possible ways and these words are written out as in dictionary, then the word $SACHIN$ appears at serial number

(A) 603 (B) 602 (C) 601 (D) 600

MPC197

2. How many words can be made from the letters of the word $DELHI$, if L comes in the middle in every word

(A) 12 (B) 24 (C) 60 (D) 6

MPC198

3. If the letters of the word "$VARUN$" are written in all possible ways and then are arranged as in a dictionary, then the rank of the word $VARUN$ is

(A) 98 (B) 99 (C) 100 (D) 101

MPC199

4. The sides AB, BC, CA of a triangle ABC have 3, 4, 5 points respectively on them. Find the number of triangle that can be constructed using these points as vertices.

(A) 105 (B) 205 (C) 305 (D) 405

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MPC200

5. The sides AB, BC, CA of a triangle ABC have 3, 4, 5 points respectively on them. Find the number of triangle that can be constructed whose two vertices lie on same side of ΔABC.

(A) 105 (B) 205 (C) 145 (D) 405

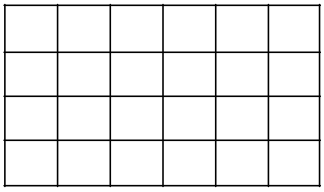
MPC201

6. The sides AB, BC, CA of a triangle ABC have 3, 4, 5 points respectively on them. Find the number of triangle that can be constructed whose vertices lies on three different sides of ΔABC.

(A) 40 (B) 60 (C) 80 (D) 100

MPC202

7. Number of rectangles in the grid shown which are not squares is



(A) 160 (B) 162 (C) 170 (D) 185

MPC203

8. In a plane there are $'4'$ circle & $'6'$ straight lines then maximum number of intersection point is

(A) 60 (B) 48 (C) 27 (D) 75

MPC204

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9. The number of ways of choosing 4 cards from a pack of 52 playing cards such that these four cards are of the same suit,

(A) 2860 (B) 495 (C) 105625 (D) 29900

MPC205

10. The number of ways of choosing 4 cards from a pack of 52 playing cards such that these four cards belong to four different suits,

(A) $13! \times 4$ (B) 13^4 (C) 4^{13} (D) 13!

MPC206

11. The number of ways of choosing 4 cards from a pack of 52 playing cards such that these are face cards,

(A) 2860 (B) 495 (C) 105625 (D) 29900

MPC207

Comprehension Type Question

Paragraph for Question 12 & 13

A cricket team of eleven players is to be selected out of 6 batsman, 5 bowlers, 2 all rounders and 2 wicket keepers. Consider an allrounder as a player who is a batsman as well as a bowler.

12. The number of selections containing exactly one wicket keeper and atleast one all rounder is –

(A) ${}^2C_1(2.\ {}^{11}C_9 + {}^{11}C_8)$ (B) ${}^2C_1.\ {}^2C_9.\ {}^{11}C_9$

(C) ${}^2C_1.\ ({}^{12}C_3 + {}^{11}C_3)$ (D) ${}^4C_2.\ {}^{11}C_9$

MPC208

13. Number of selections when a particular wicket keeper do not want to play if a particular batsman and a particular bowler are both playing, is (where it is not necessary that a wicket keeper must be a part of playing eleven)

(A) ${}^{13}C_{11} + {}^{13}C_{10} + {}^{12}C_9$ (B) ${}^{15}C_{11} - 2.\ {}^{13}C_3$

(C) ${}^{13}C_{11} + 2.\ {}^{13}C_{10} + {}^{12}C_9$ (D) ${}^{13}C_{11} + {}^{13}C_{10} + {}^{12}C_8$

MPC209

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11	12	13
Answer	C	B	C	B	C	B	A	D	A	B	B	A	C

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RACE
Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-04

Single Correct Question

1. How many words can be formed with the letters of the word $MATHEMATICS$ by rearranging them

(A) $\frac{11!}{2!2!}$ (B) $\frac{11!}{2!}$ (C) $\frac{11!}{2!2!2!}$ (D) $11!$

MPC210

2. The total number of permutations of the letters of the word "$BANANA$" is

(A) 60 (B) 120 (C) 720 (D) 24

MPC211

3. In how many ways we can arrange 3 red flowers, 4 yellow flowers and 5 white flowers in a row?

(A) 27720 (B) 50400

(C) 10,080 (D) 40,320

MPC212

4. How many words can be formed by using all the letters of the word $MISSISSIPPI$, if consonants are in alphabetical order.

(A) $\frac{11!}{3!.8!}$ (B) $\frac{11!}{4!.7!}\times 2$ (C) $\frac{11!}{4!.7!}$ (D) $\frac{11!}{4!.7!}\times\frac{1!}{2!}$

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MPC213

5. Consider word $ALLENDIGITAL$, find number of ways of arranging the letters. If relative position of vowels and consonant remain same.

(A) 25600 (B) 31600

(C) 25200 (D) 22500

MPC214

6. Find the number of arrangements of the letters of the word $INDEPENDENCE$. In how many of these arrangements, do the vowels never occur together

(A) 1646400 (B) 1846800

(C) 1546400 (D) 1606500

MPC215

7. A cricket team comprising of $A, B, C, D \ldots\ldots, K$ is to be sent for batting. If A wants to bat before J and J wants to bat after G then the number of batting orders if other players could go in any order is?

(A) 13405600 (B) 11205060

(C) 12200560 (D) 13305600

MPC216

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8. Consider word *ASSASSINATION*, find number of ways of arranging the letters. If vowels are in the same order.

(A) $\frac{13!}{(4!)(2!)(6!)}\times 3!$ (B) $\frac{13!}{(4!)(2!)(6!)}$

(C) $\frac{13!}{(5!)(2!)(5!)}\times\frac{1}{2!}$ (D) $\frac{12!}{(4!)(2!)(5!)}$

MPC217

Comprehension Type Question

Paragraph for Question 9 & 10

There are 5 boys and 5 girls.

9. If they are seated in a straight line, then the number of ways in which all boys and all girls are together is-

(A) (5!5!)2 (B) 4!5! (C) 10! (D) 7!3!

MPC218

10. Number of ways in which they can be divided into two groups, if first group contains 2 boys & 3 girls, second group contains 3 boys and 2 girls is-

(A) 50 (B) 20 (C) 10 (D) 100

MPC219

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	A	C	C	A	D	B	A	D

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-05

Single Correct Question

1. A shopkeeper having $'n'$ no. of different books, each book having $'p'$ copies, then total no. of different selections of books can be made is :

(A) $P^n - 1$ (B) $(p+2)^n - 1$

(C) $P^n + 1$ (D) $(P+1)^n - 1$

MPC220

2. Six species (having same or difference internally) $(A_1, A_2), (B, B, B), (M_1, M_2, M_3, M_4), (C, C), B, F$ then total number of selections of species or number of ways of taking atleast one specie.

(A) 2081 (B) 3071 (C) 3072 (D) 4001

MPC221

3. Find the total number of proper factors of the number 35700

(A) 69 (B) 70 (C) 71 (D) 72

MPC222

4. There are 10 questions in an examination. In how many ways can a candidate answer the questions, if he attempts atleast one question.

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(A) $2^{10} - 1$ (B) 2^{10} (C) $2^9 - 1$ (D) $2^{11} - 1$

MPC223

5. Find the number of groups that can be made from 5 red balls, 3 green balls and 4 black balls, if at least one ball of all colours is always to be included. Given that all balls are identical except colours.

(A) 30 (B) 40 (C) 50 (D) 60

MPC224

6. Find the total number of 4 letter words formed using four letters from the word $'PARALLELOPIPED'$.

(A) 2780 (B) 2480

(C) 2680 (D) 3000

MPC225

7. In how many ways 3 teams of 11 player's each, 4 team's 6 player's each, 2 teams of 15 player's each can be formed out of 87 player's.

(A) $\frac{87!}{(11!)^3(6!)^4(15!)^2}$ (B) $\frac{87!}{(11!)^3(6!)^4(15!)^2} \times \frac{1}{3!4!2!}$

(C) $\frac{87!}{3!11!4!6!2!15!}$ (D) $\frac{87!}{(11!)^3(6!)^4(15!)^2} \times \frac{9!}{3!4!2!}$

MPC226

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8. In how many ways six different books can be distributed between four persons. So that each person gets atleast one book.

(A) $\left[\frac{6!}{(1!)^3 3!2!}+\frac{6!}{(1!)^2(2!)^2(2!)^2}\right]$ (B) $\left[\frac{6!}{(1!)^3 3!3!}\right]\times 4!$

(C) $\left[\frac{6!}{(1!)^3 3!3!}+\frac{6!}{(1!)^2(2!)^2 2!2!}\right]\times 4!$ (D) $\left[\frac{6!}{(1!)^2(2!)^2 2!2!}\right]\times 4!$

MPC227

Comprehension Type Question

Paragraph for Question 9 to 11

Consider a word "*DEMONETIZATION*". New words are formed (with or without meaning) by using all the letters of the given word.

9. Number of words which contain the word "*MODI*" is

(A) $\frac{\lfloor 11}{(\lfloor 2)^3}$ (B) $\frac{\lfloor 11}{(\lfloor 2)^3 \lfloor 3}$ (C) $\frac{\lfloor 11}{(\lfloor 2)^5}$ (D) $\frac{\lfloor 11}{(\lfloor 2)^5 \lfloor 5}$

MPC228

10. The number of words so that relative order of vowels and consonants remains same is -

(A) $\frac{\lfloor 7}{\lfloor 2 \lfloor 2 \lfloor 2}$ (B) $\frac{\lfloor 7}{\lfloor 2 \lfloor 2}$ (C) $\left(\frac{\lfloor 7}{\lfloor 2 \lfloor 2 \lfloor 2}\right)\left(\frac{\lfloor 7}{\lfloor 2 \lfloor 2}\right)$ (D) $\frac{\lfloor 14}{\lfloor 7 \lfloor 7}$

MPC229

11. Number of ways 4 letters can be selected from the word "*DEMONETIZATION*" is -

(A) 276 (B) 150 (C) 126 (D) 140

MPC230

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Matrix Match Type Question

12. Let "*MAHATMAGANDHI*" be a one word, then

	Column-I	Column-II
(A)	Number of selection of 5 letters such that each selection consists of all different letters is	(P) $^7C_3$
(B)	Number of selection of 5 letters such that each selection consists of 4 alike and 1 different letter is	(Q) 3. $^7C_3$
(C)	Number of selection of 5 letters such that each selection consists of 2 alike and 3 different letters is	(R) $^7C_1$
(D)	Number of selection of 5 letters such that each selection consists of 4 different letters is	(S) $^3C_2 . {}^6C_1$ (T) $^8C_3$

MPC231

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	D	B	C	A	D	A	B	C	A	C
Question	11	12								
Answer	A	(A→T; B→R; C→Q; D→Q)								

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RACE
Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-06

Single Correct Question

1. Find the number of permutations of letters a, b, c, d, e, f, g taken all at a time if neither $'beg'$ nor $'cad'$ pattern appear.
(A) $7! + 5! + 5! + 3!$ (B) $7! - 5! - 5! - 3!$
(C) $7! - 5! + 5! - 3!$ (D) $7! - 5! - 5! + 3!$

MPC232

2. Find the number of n digit numbers formed using first 5 natural numbers, which contain the digits 2 & 4 essentially.
(A) $5^n - 4^n - 4^n + 3^n$ (B) $5^n + 4^n + 4^n + 3^n$
(C) $5^n + 4^n - 4^n + 3^n$ (D) $5^n - 4^n - 4^n - 3^n$

MPC233

3. How many different words can be formed by using $2 - A's, 2 - B's, 2 - C's, D, E, F, G$ so that no two alike letters together.
(A) 236880 (B) 144000 (C) 250104 (D) 336800

MPC234

4. Find the number of ways in which the letters of the word $'KUTKUT'$ can be arranged so that no two alike letters are together.
(A) 30 (B) 35 (C) 25 (D) 40

MPC235

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5. In how many ways can we arrange 6 different flowers in a circle?
(A) 110 (B) 120 (C) 130 (D) 140

MPC236

6. In how many ways we can form a garland using 6 different flowers?
(A) 30 (B) 40 (C) 50 (D) 60

MPC237

7. The number of ways in which 20 different pearls of two colours can be set alternately on a necklace, there being 10 pearls of each colour, is
(A) $9! \times 10!$ (B) $5(9!)^2$ (C) $(9!)^2$ (D) None of these

MPC238

8. A person invites a group of 10 friends at dinner. They sit 5 on one round table and 5 on other round table,
(A) $\frac{10!}{25}$ (B) $\frac{11!}{25}$ (C) $\frac{12!}{24}$ (D) $\frac{13!}{26}$

MPC239

9. The number of ways in which 6 men and 5 women can dine at a round table if no two women are to sit together is given by
(A) $6! \times 5!$ (B) 30 (C) $5! \times 4!$ (D) $7! \times 5!$

MPC240

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9
Answer	D	A	A	A	B	D	B	A	A

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RACE Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-07

Single Correct Question

1. Find number of ways in which 30 identical mangos can be distributed among 5 persons.

(A) ${}^{31}C_4$ (B) ${}^{32}C_4$

(C) ${}^{33}C_4$ (D) ${}^{34}C_4$

MPC241

2. Total number of ways of distributing 7 identical computers to 3 persons, so that each receive atleast one computer is

(A) ${}^{5}C_2$ (B) ${}^{6}C_2$

(C) ${}^{7}C_2$ (D) ${}^{8}C_2$

MPC242

3. A shelf contains 8 separate compartments. Find number of ways in which 15 identical marbles can be placed in the compartment so that no compartment is empty.

(A) ${}^{14}C_6$ (B) ${}^{14}C_7$

(C) ${}^{16}C_8$ (D) ${}^{22}C_7$

MPC243

4. Number of ways in which 16 identical toys are to be distributed among 3 children such that each child does not receive less than 3 toys will be

(A) 96 (B) 16

(C) 36 (D) 46

MPC244

5. Aman has to buy 25 mangoes in four different variety buying at least 4 of each variety. In how many ways can he plan his purchases; if mangoes of each variety are identical and available in abundance.

(A) ${}^{12}C_3$ (B) ${}^{13}C_3$

(C) ${}^{11}C_2$ (D) ${}^{12}C_4$

MPC245

6. Number of non-negative integral solution of the inequality $x + y + z + t \leq 30$.

(A) ${}^{31}C_4$ (B) ${}^{32}C_4$

(C) ${}^{33}C_4$ (D) ${}^{34}C_4$

MPC246

7. For an equation $x + y + z = 20$, then find number of natural solution

(A) ${}^{12}C_2$ (B) ${}^{18}C_2$

(C) ${}^{22}C_2$ (D) ${}^{19}C_2$

MPC247

8. For an equation $x + y + z = 20$, then find number of integral solution if $x \geq 1, y \geq 3, z \geq 4$

(A) ${}^{14}C_2$ (B) ${}^{18}C_2$

(C) ${}^{22}C_2$ (D) ${}^{19}C_2$

MPC248

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Matching List Type Question

9. Match List-I with List-II and select the correct answer using the code given below the list.

	List-I	**List-II**
(P)	Number of outcomes in the case of 6 distinct dice	(1) 27216
(Q)	Number of outcomes in the case of 6 alike dice	(2) 78
(R)	Number of outcomes in case of 2 alike and 4 distinct dice	(3) 462
(S)	Number of terms in the expansion of $(x+y+z)^{11}$	(4) 46656

Codes :

	P	**Q**	**R**	**S**
(A)	1	3	2	3
(B)	4	3	1	3
(C)	1	2	4	2
(D)	4	3	1	2

MPC249

Integer Type Question

10. In Mahabharat battle, Kaurav sena made a Chakravyuh by placing 12 warriors on a circle. To break this, Abhimanyu had to kill 5 warriors such that no two warriors are consecutive. If number of ways in which Abhimanyu could do this was N^2, then the value of N is

MPC250

11. If N is number of arrangement of the letters of the word RAMARAAM in which no two alike letters are together, then the unit digit of N is

MPC251

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	D	B	B	C	A	D	D	A	D	6	4

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RACE Regular Analysis Through Continuous Exercise

Permutations and Combinations: RACE-08

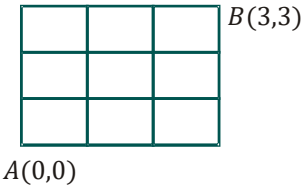
Single Correct Question

1. The number of ways in which an insect can move from left bottom corner of a chess board to the right top corner, if it is given that it can move only upside or right, along the lines is-

(A) ${}^8C_4$ (B) ${}^{16}C_8$ (C) ${}^{15}C_8$ (D) ${}^8C_2$

MPC252

2. In how many ways can a person go from point A to point B if he can travel only to the right or upward along the lines (Grid Problem)?



(A) 20 (B) 30 (C) 40 (D) 50

MPC253

3. Find sum of all numbers formed using the digits 2,4,6,8 taken all at a time and no digit being repeated.

(A) 111220 (B) 121220 (C) 133320 (D) 141110

MPC254

4. Find the sum of all the numbers greater than 1000 using the digits 0,1,2,2.

(A) 16110 (B) 16111 (C) 17230 (D) 15410

MPC255

5. A person writes letters to his 5 friends and addresses the corresponding envelopes. Number of ways in which the letters can be placed in the envelope, so that atleast two of them are in the wrong envelopes, is

(A) 1 (B) 2 (C) 118 (D) 119

MPC256

6. A match the column type question consists of 5 parts and column A has 5 questions and column B has 5 answers and each question having single correct and different (one on one) matching answers then in how many ways the student can mark all 5 wrong answers.

(A) 14 (B) 34 (C) 44 (D) 56

MPC257

7. A person writes letters to six friends and addresses the corresponding envelopes. In how many ways can the letters be placed in the envelopes so that atleast two of them are in the wrong envelopes.

(A) 619 (B) 719 (C) 819 (D) 919

MPC258

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8. Find the number of positive integral solution of the equation $x.y.z = 120$.

(A) 70 (B) 75 (C) 80 (D) 90

MPC259

9. Find the number of positive integral solutions of the inequation $x + y + z \geq 150$, where $0 < x \leq 60, 0 < y \leq 60, 0 < z \leq 60$.

(A) ${}^{33}C_3$ (B) ${}^{34}C_3$ (C) ${}^{35}C_5$ (D) ${}^{36}C_6$

MPC260

10. In how many ways can 12 identical apples be distributed among four children if each gets atleast 1 apple and not more than 4 apples.

(A) 25 (B) 27 (C) 29 (D) 31

MPC261

11. Set $A = \{1, 2, 3, 4, 5, 6, 7\}$ and $B \subseteq A$ & $C \subseteq A$. In how many ways sets B & C can be formed such that $B \cap C = \phi$?

(A) 3^7 (B) 3^4 (C) 2^5 (D) $3^5 - 1$

MPC262

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	B	A	C	A	D	C	B	D	A	D	A

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